

PAIMANA SENTINEL

Complete End-to-End Project Report

A simple but detailed guide to the data pipeline, delay prediction model, risk explanations, and Phase 2 backend

Current project state

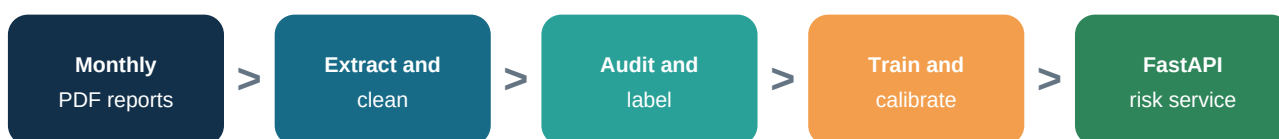
Dataset v1.0: frozen | Delay Model v1.1: frozen | Phase 2 backend: complete

Prepared from the validated local project artifacts

1. The project in one minute

PAIMANA Sentinel is an early-warning system. It estimates whether an infrastructure project's reported completion target is likely to move later during the next three calendar months.

Imagine an officer monitoring hundreds of roads, rail lines, airports, power projects, and other large public works. Reading every report manually is slow. PAIMANA Sentinel organizes the monthly records, checks their quality, learns historical warning patterns, and turns the latest project snapshot into a risk probability and an understandable explanation.



Every stage keeps provenance and avoids inventing unknown values.

22,972	2,804	99.88%	3 months
clean project-month records	unique projects	target-date coverage	prediction horizon

What it predicts: the probability that the currently reported expected or revised completion target will move to a later date within the next three months.

What it does not predict: it does not prove why a delay happened, replace engineering judgment, approve expenditure, or make a final government decision.

Simple example: if a project's current information suggests an estimated 64% chance of a target-date extension in the next three months, the system may show HIGH or CRITICAL risk according to frozen validation-selected thresholds. It also shows which current facts pushed risk up or down and how reliable the input data was.

2. Why PAIMANA Sentinel is needed

Large infrastructure projects are reported repeatedly over time. A single monthly report may show cost, expenditure, physical progress, target dates, ministry, sector, and location. The most useful warning is not simply 'this project is already delayed.' The useful question is: **does its current pattern suggest that the reported target may be pushed back soon?**

The project therefore changes monthly reporting data into a forward-looking monitoring signal. This can help teams prioritize review effort. A high-risk project can be examined sooner, while a low-risk project still remains visible in the monitoring system.

The primary use cases are:

- Portfolio triage: rank projects for attention without hiding the underlying facts.
- Project review preparation: show the risk trajectory over several months, not only today's score.
- Transparent warning: explain which reported features were associated with higher or lower risk.
- Data-quality awareness: keep the delay probability separate from confidence in the input data.

- Auditability: retain source provenance, anomalies, extraction audits, label audits, and leakage checks.

The intended user is a monitoring officer, analyst, project manager, or decision-support team. The design is also suitable for an SIH demonstration because it connects document extraction, data engineering, machine learning, explainability, and a service API in one defensible workflow.

3. End-to-end architecture

The system has four layers. Each layer has a different job and can be inspected independently.

Layer	Input	What happens	Output
1. Source and extraction	Cached monthly official PDF reports	Find all ongoing project rows; preserve page, raw row evidence, serial number, and codes	Monthly raw snapshots
2. Dataset engineering	Extracted snapshots	Normalize text and dates; repair stable metadata; create quality flags; link project identities	Frozen Dataset v1.0
3. Model training	Known 3-month delay-label rows	Use leakage-safe current/past features; temporal train/validation/test; calibrate and explain	Frozen CatBoost Model v1.1
4. Serving backend	One snapshot or a project ID	Validate input; score; compute local SHAP; generate confidence and history	FastAPI JSON responses

The dataset and model are frozen. The backend reads them; it does not rewrite them or retrain automatically. This separation prevents an API request from silently changing the evidence or model.

A request follows this path:

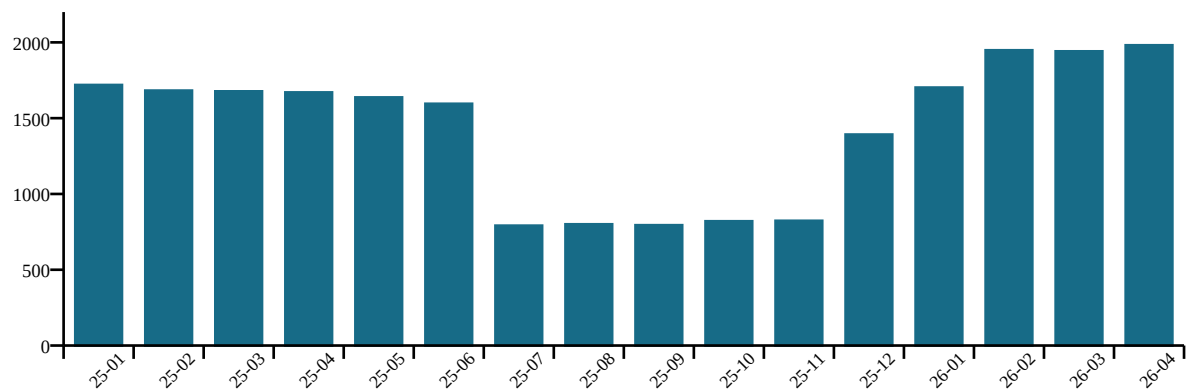
- Pydantic checks that required feature fields exist and rejects malformed values.
- The already-loaded CatBoost model generates a raw probability.
- The frozen isotonic calibrator adjusts that value so it behaves more like a probability.
- Frozen thresholds map the probability to a risk band.
- Local SHAP values identify the strongest risk-increasing and risk-reducing drivers.
- A separate reliability calculation checks data quality, missing values, identity source, and extreme inputs.

4. Source reports and reporting regimes

The project uses cached monthly PDFs only. Reports cover January 2025 through April 2026 in the frozen dataset. The source format changed during this period, so the pipeline records an explicit reporting regime instead of pretending every report has an identical schema.

Period	Reporting regime	Meaning
Jan-Jun 2025	OCMS_LEGACY	Older reporting layout and identifier/date semantics
Jul-Aug 2025	PAIMANA_TRANSITION	Transition layout; special extraction and semantic checks
Sep 2025 onward	PAIMANA	Newer PAIMANA reporting layout

Clean project-month rows by reporting month



The Jul-Nov 2025 drop reflects a real reporting-scope transition, not forced parser loss.

The lower Jul-Nov 2025 counts were investigated. The official declarations themselves listed about 791 to 823 projects, so the decline from roughly 1,600 was mainly a real scope change. The builder did not invent rows or force these months to resemble the older OCMS scope.

5. Extraction validation

The transition months received a row-by-row reconciliation. The audit checks declared project count, parsed rows, unique serial numbers, missing serials, duplicate serials, duplicate project codes, duplicate canonical identities, and rows removed by deduplication.

Month	Declared	Parsed	Unique serials	Completeness	Status
2025-07	791	791	791	100.0%	COMPLETE
2025-08	800	800	800	100.0%	COMPLETE
2025-09	794	794	794	100.0%	COMPLETE
2025-10	820	820	820	100.0%	COMPLETE
2025-11	823	823	823	100.0%	COMPLETE

All five audited transition/PAIMANA months reached their source-declared serial counts with no unresolved rows. This means 100% completeness is supported by source evidence for these specific months, not guessed from surrounding months.

If two rows had shared an uncertain parsed project code but had different serial numbers, names, or raw project cells, the intended rule was to retain both and flag a possible code collision. Malformed but identifiable rows were to be retained at lower identity confidence rather than silently deleted.

6. Cleaning and identity linking

A project appears once per reporting month. The builder creates a canonical project identity so history can be followed across reports. Identity may come from a PAIMANA project code, a legacy OCMS code, or a carefully marked fallback identity when no reliable official code is available.

85.51%	14.06%	0.43%	17
rows using PAIMANA identity	rows using legacy identity	rows using fallback identity	ambiguous identities

Cleaning rules include:

- Parse YYYY-MM-DD, DD-MM-YYYY, DD/MM/YYYY, and supported month-year date forms.
- Choose the current target from anticipated date first, then revised date, then original date; otherwise keep it missing.
- Normalize embedded newlines, tabs, and repeated whitespace in project name, sector, state, ministry/department, and agency.
- Repair stable metadata using the most frequent normalized value; use completeness/length and lexical order only as deterministic tie-breakers.
- Never forward-fill changing numeric fields such as expenditure, progress, or cost.
- Preserve anomalous source rows and add flags instead of deleting inconvenient observations.

The final frozen dataset has **zero duplicate project-month records**. That matters because duplicated monthly observations could over-weight a project during training and distort trends.

7. Dataset v1.0 at a glance

22,972	2,804	90.35	0
clean observations	projects	average quality score	duplicate project-months

History available	Projects
At least 3 months	2,609
At least 6 months	1,868
At least 12 months	501

A 'project-month' is one project's state in one month. Therefore, 22,972 observations do not mean 22,972 projects. They are repeated snapshots of 2,804 projects, which is what makes trajectory features and future labels possible.

Important quality findings were retained: 301 physical-progress decreases, 938 cumulative-expenditure decreases, 85 suspicious resets, one physical-progress value above 100%, 2,933 target-source transitions, and 4,028 possible scope-change flags. These flags are evidence for review, not automatic proof that the source is wrong.

8. What the delay label means

The main target is `delay_event_next_3m`. It asks whether the reported completion target moves later during the next three exact calendar months.

For a project snapshot at month t , a positive label (1) is created only when all of the following are true:

- The target date at t is known.
- The exact months $t+1$, $t+2$, and $t+3$ all exist; no gaps are allowed.
- Target meanings remain comparable across the reporting regime/source transition.
- At least one comparable target in those three months moves later than the previous comparable target.

A negative label (0) is used only when future coverage is complete, dates are known, semantics are comparable, and no later movement happens. If any condition is unknown, the label stays blank (NaN). Unknown is never converted into a convenient negative.

Target	Usable	Positive	Negative	Positive rate
delay 3m	10,368	1,933	8,435	18.64%
delay 6m	3,733	752	2,981	20.14%
cost 3m	11,609	189	11,420	1.63%
cost 6m	4,114	133	3,981	3.23%

Only the **delay 3m** target is used by the current model. Six-month delay and cost datasets exist as audited outputs, but the present project phase deliberately does not train or serve those models.

9. Target semantics and transition safeguards

Older OCMS reports and newer PAIMANA reports do not always expose completion dates in the same way. For example, an anticipated date could disappear when moving to a report that exposes only revised/original dates. A naive comparison could call this a new delay even when the later date had already been known.

The pipeline records `current_target_source` as `ANTICIPATED`, `REVISED`, `ORIGINAL`, or `MISSING`. It audits changes around June-July-August 2025, marks `TARGET_SOURCE_TRANSITION`, and excludes semantically uncertain future comparisons from labels.

2,933	4,463	NaN	0
target-source transitions	delay exclusions for ambiguity	value kept for unknown labels	fabricated dates or rows

Defensibility rule: fewer trustworthy labels are better than many labels created by treating missing or incomparable information as 'no delay.'

10. Leakage prevention and temporal testing

Data leakage happens when a model accidentally sees information from the future. That can make offline scores look excellent while the real system fails. PAIMANA Sentinel uses an explicit feature registry and leakage guard.

- Features at month `t` use only month `t` and exact previous months.
- Target columns, future coverage flags, validity fields, raw source text, identifiers, and reporting regime are excluded as predictors.
- Lag features do not substitute a later observation when an expected prior month is missing.
- The main evaluation uses time order, not a random row-level split.
- Projects may repeat across time because deployment will monitor known projects again; a separate unseen-project check measures robustness.

Partition	Period	Rows	Projects	Positive rate
Train	2025-01 to 2025-08	6,065	1,786	16.29%
Validation	2025-09 to 2025-11	2,148	774	15.64%
Frozen test	2025-12 to 2026-01	2,155	1,313	28.26%

Model V1.1 used September-October 2025 for selection and November 2025 for calibration. The frozen December 2025-January 2026 test was evaluated only after model, calibration, and thresholds were fixed.

11. Models that were compared

Four types of classifier were considered in Model V1: a Dummy baseline, Logistic Regression, CatBoost, and XGBoost. The dummy shows what happens without useful learning. Logistic Regression is a transparent conventional benchmark. CatBoost and XGBoost can learn nonlinear interactions from tabular data.

Model	Why it matters	V1 test PR-AUC
Dummy prior	No real discrimination; probability follows class prevalence	0.2826
Logistic Regression	Transparent statistical benchmark	0.5608
CatBoost V1	Handles nonlinear tabular/categorical patterns	0.5527
XGBoost V1	Alternative boosted-tree benchmark	0.5649

The first CatBoost model was kept at MODEL_STATUS=YELLOW because it had useful signal but did not clearly beat Logistic Regression and calibration error remained high. This honest result triggered one focused feature/calibration improvement pass rather than threshold manipulation.

Model V1.1 added a small leakage-safe set: progress acceleration, months since meaningful progress, recent target-revision count and recency, expenditure velocity versus progress velocity, progress required per remaining month, and peer-relative progress. CatBoost_2 was selected using validation data only and calibrated with isotonic regression.

12. Frozen Model V1.1 performance

0.606	0.821	0.644	0.427
test PR-AUC	test ROC-AUC	precision	recall

Metric	Value	Plain-language meaning
PR-AUC	0.6061	Ability to find minority delay events while controlling false alerts; primary ranking metric
ROC-AUC	0.8214	Chance that a random delayed case is ranked above a random non-delayed case
Precision	0.6436	Among generated alerts, the share that became positive delay events
Recall	0.4269	Share of actual delay events caught by the selected threshold
F1	0.5133	Single balance of precision and recall
Brier score	0.1848	Probability error; lower is better
ECE	0.1531	Average calibration gap; lower is better

At the frozen balanced threshold of 0.225, the test set produced 260 true positives, 144 false positives, 349 false negatives, and 1402 true negatives. It generated 404 alerts across 2,155 test rows.

The model is useful but not perfect. It is still marked **YELLOW**, mainly because calibration error remains material and recall on completely unseen projects is low. This means it is suitable as a monitored decision-support prototype, not an automatic authority.

13. Generalization and temporal stability

Evaluation slice	PR-AUC	ROC-AUC	Precision	Recall	Brier	ECE
Full frozen test	0.606	0.821	0.644	0.427	0.185	0.153

Evaluation slice	PR-AUC	ROC-AUC	Precision	Recall	Brier	ECE
Unseen projects	0.532	0.793	0.517	0.147	0.261	0.262
High-quality subset	0.590	-	-	-	-	-

Test month	Rows represented	PR-AUC	ROC-AUC	Precision	Recall	Brier	ECE
2025-12	865	0.602	0.838	0.630	0.542	0.134	0.095
2026-01	1,290	0.618	0.812	0.653	0.374	0.219	0.193

The model ranks unseen projects reasonably, but recall falls to about 14.7%. In plain language, it is conservative on projects it has never observed. The high-quality subset PR-AUC is about 0.590, supporting the separate confidence layer and the value of better source data.

A good overall score does not guarantee equal performance in every month, sector, state, or project type. Monitoring must continue after deployment.

14. The most important Model V1.1 features

Feature	Simple explanation
months_to_target	How many months remain until the currently reported target date
required_progress_per_remaining_month	Physical progress still needed divided by remaining months
ministry_department	Responsible ministry/department; used as a pattern, not a causal claim
current_time_slip_months	How far the current target has moved relative to the original target
schedule_pressure	How demanding the remaining schedule looks given progress and time
state	Location category used for pattern learning and subgroup review
peer_relative_progress_pct	Progress compared with similar projects in a leakage-safe contemporaneous peer group
project_age_months	Time elapsed since project start/approval
physical_progress_pct	Latest reported physical completion percentage
spend_ratio_pct	Cumulative expenditure relative to current forecast cost

These are global top features, meaning they were influential across many predictions. For a particular project, local SHAP can show a different order. A category such as state or ministry must never be interpreted as proof that the location or organization caused a delay.

Other allowed feature families include progress/spend changes over 1, 3, and 6 months; progress velocity; stagnation streak; cumulative target and cost revisions; data-quality score; missing-core-field count; progress acceleration; time since meaningful progress; recent target revision recency; expenditure/progress velocity gap; and current cost overrun.

15. Probability, risk level, and confidence are different

The API deliberately returns three related but separate ideas:

Output	Question answered	Example
Delay probability	How likely is a reported target-date move in the next 3 months?	0.61 means an estimated 61% risk
Risk level	Which frozen operational band contains that probability?	LOW, MEDIUM, HIGH, or CRITICAL
Prediction confidence	How trustworthy/complete is the input row for scoring?	HIGH, MEDIUM, or LOW plus a 0-100 score

Frozen V1.1 thresholds are: EARLY_WARNING 0.090, BALANCED 0.225, CONSERVATIVE 0.350, and CRITICAL 0.350. The risk-level helper uses the frozen category mapping: below BALANCED is LOW; BALANCED to CONSERVATIVE is MEDIUM; CONSERVATIVE to CRITICAL is HIGH; CRITICAL and above is CRITICAL. Because CONSERVATIVE and CRITICAL are equal in this artifact, the HIGH interval is empty. This unusual but valid result is preserved rather than inventing a new threshold from test data.

Confidence is calculated mainly from data_quality_score, missing model inputs, fallback/generated identity, and numeric features outside the central training range. It is **not** the model probability. A project can have high delay risk but low confidence if its source data is incomplete or unusual.

16. How explainability works

The backend uses local SHAP values from the frozen CatBoost model. SHAP starts from the model's baseline prediction and attributes movement up or down to the current row's feature values.

- Top risk drivers are features with the strongest positive contribution for that prediction.
- Top protective drivers are features with the strongest negative contribution.
- The API returns up to five of each with plain-language labels and observed values.
- Global SHAP is not recomputed on every request; only the requested local explanation is calculated.
- The explanation says these are predictive associations, not causal effects.

Example wording:

Risk is primarily increased by progress required per remaining month and schedule pressure, while current physical progress and expenditure velocity reduce the estimate. These are predictive associations, not causal effects.

SHAP explains what the model used, not what action will definitely reduce delay. A future what-if simulator would require stricter non-causal language and scenario constraints; it is intentionally outside the present phase.

17. Phase 2 FastAPI backend

The backend is under backend/. The V1.1 artifact is loaded once during FastAPI startup. Project history is built from frozen cleaned/master CSV files. No endpoint writes to the dataset or retrains the model.

Module	Responsibility
main.py	Application startup and HTTP routes
schemas.py	Pydantic input/output contracts and validation
model_service.py	Frozen bundle load, prediction, calibration, risk level, local SHAP
project_service.py	Read-only project metadata, latest rows, history, cached list risks
trajectory_service.py	Chronological month-by-month scoring
explanation_service.py	Human-readable risk/protective drivers
confidence_service.py	Data reliability score separate from risk probability
config.py	Stable local artifact and dataset paths

Run command:

```
uvicorn backend.main:app --reload
```

The automatic documentation is available at <http://127.0.0.1:8000/docs> while the server is running locally.

18. API endpoint reference

Endpoint	Purpose	Main response
GET /health	Check service and frozen model	status, model_loaded, version, model status, API version
POST /predict	Score one validated snapshot	probability, risk, drivers, quality, confidence, model version
GET /projects	List available monitored projects	ID, name, sector, state, latest month, latest risk
GET /projects/{id}	Detailed project view	metadata, latest snapshot, history, current risk, explanation
GET /projects/{id}/trajectory	Historical risk trend	month, probability, risk level, quality
GET /projects/{id}/explanation	Latest local explanation	top five risk/protective factors and sentence

POST /predict requires the complete frozen model feature schema. Values may be null where the model accepts missing information, but required keys must be present. Extra unexpected fields are rejected. Numeric infinity and invalid quality scores are rejected with clear HTTP 422 validation errors.

19. Example prediction response

```
{
  "delay_probability_3m": 0.61,
  "risk_level": "CRITICAL",
  "threshold_profile": "BALANCED",
  "top_risk_drivers": [
    {"feature": "schedule_pressure", "label": "Schedule pressure", "observed_value": 1.8}
  ],
  "top_protective_drivers": [
    {"feature": "physical_progress_pct", "label": "Current physical progress", "observed_value": 82.0}
  ],
  "data_quality_score": 91.0,
  "prediction_confidence": {
    "level": "HIGH", "score": 91.4,
    "missing_feature_count": 0,
    "out_of_range_feature_count": 0,
    "fallback_identity": false
  },
  "model_version": "delay_3m_v1_1"
}
```

This is an illustrative structure, not a claim about a specific real project. The returned number means model-estimated risk, not certainty. Officers should inspect the latest source record, trajectory, data quality, and explanation before acting.

20. Risk trajectory

The trajectory endpoint scores every historical snapshot for which the frozen model features can be assembled. Results are sorted chronologically and include snapshot month, predicted three-month delay probability, risk level, and data-quality score.

A trajectory is more useful than a single score because it can answer questions such as:

- Is risk rising for three consecutive months?
- Did risk jump after progress stagnated or the schedule became compressed?
- Is a risk change accompanied by lower data quality or a reporting transition?
- Did risk fall after the latest reported progress improved?

Trajectory values are historical model scores generated using each month's available features. They should not be confused with actual observed outcomes for those same months.

A future frontend can plot this sequence, but the current phase deliberately stops at the backend API.

21. Data quality, safety, and auditability

The system keeps quality and provenance visible throughout the workflow. Key flags include cumulative-expenditure decrease, physical-progress decrease, progress above 100%, negative cost, missing target date, suspicious cost/progress reset, metadata inconsistency, possible partial extraction, stale update, reporting-regime change, target/cost source transition, possible code collision, and possible scope change.

Quality flags have INFO, WARNING, or CRITICAL severity. Obvious impossible values are strongly flagged, but the pipeline does not automatically alter them without deterministic source evidence.

Backend safeguards include:

- Read-only use of frozen datasets and model artifacts.
- One model load at startup rather than reloading/retraining per request.
- Strict schema validation and bounded quality score.
- Deterministic predictions for the same input.
- No direct project ID as a predictive feature.
- No reporting regime as a Model V1.1 predictor; it is for diagnostics only.
- Local explanation and confidence provided alongside the risk score.

22. Testing evidence

The Phase 2 backend completed eight automated tests:

- Health endpoint and frozen model load.
- Prediction probability remains between 0 and 1.
- Risk and confidence levels use allowed values.
- Same input produces the same response.
- Malformed prediction requests are rejected.
- Project listing and lookup work.
- Trajectory is chronologically ordered.
- Explanation output is present and explicitly non-causal; unknown projects return 404.

8 / 8	36.0 ms	HTTP 200	GREEN
backend tests passed	average warm /predict latency	all valid route smoke checks	Phase 2 backend status

Latency was measured locally across 25 warm prediction requests including local SHAP explanation. It is an environment-specific engineering measurement, not a guaranteed production service-level objective.

23. Current status: what is ready and what is not

Component	Status	Interpretation
Dataset v1.0	YELLOW	Auditable and usable, but reporting transitions and semantic exclusions remain documented
Model V1	YELLOW	Useful signal; did not clearly beat the transparent baseline and calibration needed work
Model V1.1	YELLOW	Improved future-period discrimination, but calibration and unseen-project recall still need monitoring
Phase 2 backend	GREEN	Frozen model is served correctly; endpoints, trajectories, explanations, confidence, and tests work

GREEN for the backend does not turn the YELLOW model into a fully autonomous production decision-maker. It means the software serving layer is functioning correctly around the frozen model.

24. Important limitations

- The target measures a future movement in the *reported completion target*, not every possible meaning of real-world delay.
- The data comes from monthly reports and inherits their omissions, revisions, and reporting practices.
- The model finds associations; it cannot prove causes or guarantee that an intervention will work.
- Unseen-project recall is low, so new projects require caution and human review.
- Calibration varies by test month; a 60% score should not be treated as perfectly calibrated in every subgroup.
- Some categories may reflect institutional/reporting patterns and need fairness and stability monitoring.
- The present backend has no authentication, authorization, database, audit log, rate limit, container deployment, or frontend yet.
- Cost and six-month models are intentionally not trained or served in this phase.

Recommended operating rule: use PAIMANA Sentinel to decide what deserves closer review, never as the sole basis for sanctions, funding changes, or contractual action.

25. Recommended next steps

The next engineering and governance work should be staged, not rushed:

Priority	Next step	Why
1	Backend hardening	Add authentication, authorization, structured logs, request IDs, audit records, secure configuration, and deployment monitoring
2	Model monitoring	Track drift, calibration, alert volume, monthly metrics, subgroup metrics, and unseen-project behavior
3	Human review workflow	Define who reviews alerts, what evidence is shown, and how outcomes/overrides are recorded
4	Frontend	Build project search, risk cards, trajectory chart, explanations, data-quality warnings, and source links
5	Model iteration	Improve calibration and unseen-project recall using future data without changing the frozen test history
6	Peer and what-if research	Add carefully designed peer cohorts and constrained scenarios without causal claims

Any future retraining should create a new dataset/model version, preserve the old artifacts, use a newer chronological test period, and publish a change log.

26. Glossary for non-technical readers

Term	Meaning
Snapshot month	The month in which a project's status was reported
Project-month	One project observed in one month
Feature	A current or historical fact given to the model
Label / target	The future event the model learns to predict
PR-AUC	A ranking score useful when positive events are less common
ROC-AUC	A ranking score comparing positive and negative cases
Precision	How many alerts were correct
Recall	How many real positive events were found
Calibration	How closely predicted probabilities match observed frequencies
Brier score	Average squared probability error; lower is better
ECE	Summary of probability calibration gaps; lower is better
SHAP	A method that attributes a model prediction to its input features
Temporal split	Training on earlier months and testing on later months
Leakage	Accidentally using future information during prediction
Canonical project ID	A stable identity used to link the same project across reports

27. Project artifact map

Area	Important artifacts
Dataset builder	build_paimana_dataset_v3.py (preserved separately from the older fixed builder)
Frozen clean data	paimana_dataset_v3/raw_snapshots_clean.csv and training_master.csv
Primary training data	paimana_dataset_v3/training_delay_3m.csv
Dataset audits	extraction_audit.csv, target_semantics_audit.csv, identity_audit.csv, label_audit.csv, quality_flags.csv, leakage_audit.md
Model training	ml/train_delay_3m.py and ml/train_delay_3m_v1_1.py
Frozen model	ml/artifacts/models/delay_3m_v1_1_bundle.joblib
Model evidence	ml/artifacts/reports/v1_1_result.json and related evaluation reports
Backend	backend/main.py and the six supporting services/schemas/config modules
Backend tests	backend/tests/test_api.py

This report was generated from the current local frozen artifacts. It does not claim that a frontend, deployment platform, cost model, six-month model, or causal what-if engine exists.

28. Final plain-language conclusion

PAIMANA Sentinel turns difficult monthly infrastructure reports into a structured early-warning workflow. It first protects data quality: dates are parsed carefully, changing reporting regimes are marked, unknown outcomes remain unknown, and source anomalies are retained. It then trains a three-month delay classifier using only information that would have been available at the time. Finally, a FastAPI backend makes the frozen model usable through project search, predictions, history, risk trajectories, explanations, and a separate confidence score.

The project has demonstrated real predictive signal and a working serving layer. The strongest current result is the end-to-end audited engineering system, not a claim of perfect prediction. Model V1.1 remains YELLOW because calibration and unseen-project recall need improvement. The Phase 2 backend is GREEN because it reliably serves the frozen model and passed its tests.

Best one-sentence description: PAIMANA Sentinel helps monitoring teams notice which infrastructure projects may soon have their reported completion targets pushed back, explains the warning, and shows how much to trust the underlying data.

How to explain it to a friend in 60 seconds

Every month, the government reports how large infrastructure projects are progressing. PAIMANA Sentinel reads the structured records created from those reports and looks for warning patterns. It asks whether the project's completion target is likely to be moved later during the next three months. It then gives a probability, a simple risk band, the main reasons behind the estimate, and a separate score showing whether the input data is reliable. The model is useful for deciding which projects deserve attention first, but a human must still review the evidence.

Common questions

Question	Short answer
Does it know that a project will definitely be delayed?	No. It estimates probability from reported patterns.
Does a high-risk feature cause delay?	Not necessarily. SHAP shows model association, not causation.
Can it replace officers or engineers?	No. It supports prioritization and investigation.
Is the current software complete?	The dataset, frozen model, and backend exist; frontend and production hardening are future work.
Why is the model YELLOW?	Signal is useful, but calibration and new-project recall still need improvement.

End of report